

1. Water Jug Problem – Solution

The Water Jug Problem is solved by filling, emptying, and pouring water between the 4-gallon and 3-gallon jugs until exactly 2 gallons are obtained in the 4-gallon jug.

2. Mars Rover Agent – Solution

The Mars Rover acts as a goal-based agent. It senses the environment, detects obstacles, chooses a safe path, and moves toward the destination until the goal is reached.

3. 8-Queens Problem – Solution

The 8-Queens Problem is solved using the Backtracking Algorithm. Queens are placed one by one in safe positions. If a conflict occurs, the algorithm backtracks and tries another position until all eight queens are placed successfully.

4. Problem-Solving Agent – Solution

The problem-solving agent receives a goal as input, analyzes it, selects the appropriate action using predefined rules, and performs the action to achieve the goal.

5. Uniform Cost Search (UCS) – Solution

The Uniform Cost Search algorithm explores the graph by expanding the node with the lowest cumulative cost first. It continues until the goal node is reached, ensuring that the path found has the minimum total cost.